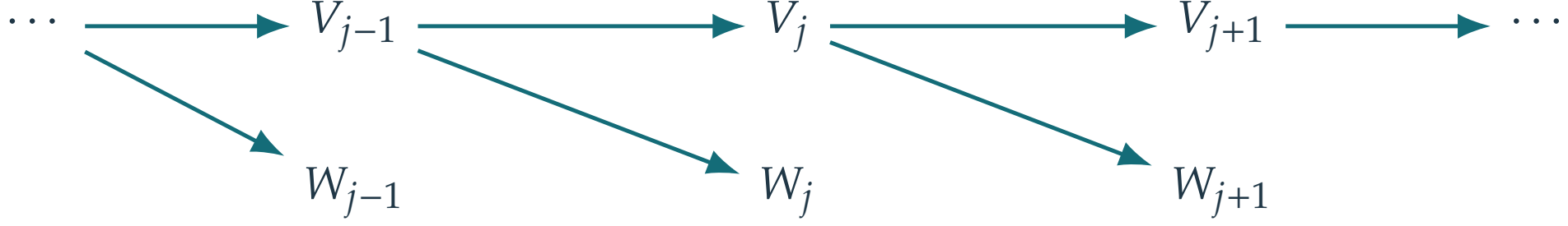

$$V_{j-1} = V_j \oplus^{\perp} W_j$$